Win Nodes for Structured Gaussian Bayes Nets Admitting Deterministic Order-Statistic Queries

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First version: September 1, 2026.
This version: November 9, 2026.

Abstract

Order-statistic queries and evidence, such as which of several correlated variables is largest or who won a contest, have long resisted exact treatment in Bayes nets. For correlated Gaussian variables the probability that one is the maximum is an orthant probability, which Bayesian optimization and reinforcement learning call intractable and approximate, and which rating systems reach only through expectation propagation. We show that when the dependence among Gaussian variables runs through a few shared factors, or along a Markov chain, the probability that each variable is the largest can be computed deterministically at a cost linear in the number of variables. In the factor case, which rests on a recent algorithm for correlated contests, the same computation gives the derivatives of these probabilities and the likelihood of an observed winner. Under a posterior, the probabilities are the action probabilities of Thompson sampling.

1 Introduction

The probability that one of several correlated Gaussian quantities is the largest has a reputation for being intractable. Tarbouriech et al. (2023) call it the key object of Bayesian reinforcement learning and say that computing it "involves computing several complicated integrals with respect to the posterior and is intractable in most cases"; their Section 4 opens with "computing this probability is intractable in general". Hennig and Schuler (2012) need the same probability over the candidate points of Bayesian optimization. They too call it intractable, citing Plackett (1954), and approximate it by expectation propagation at a cost of $O(n^4)$, which limits them to about fifty candidates.

Using an observed ranking or winner as evidence has also relied on approximation. TrueSkill (Herbrich et al., 2007) writes the likelihood of a multi-player ranking as a product of pairwise comparisons between adjacent finishers. That factorization is exact, but the posterior it produces is not Gaussian, and TrueSkill approximates it by expectation propagation. The general-purpose alternative is simulation. GHK is the standard sampler for Gaussian orthant probabilities, and Ridgway (2016) shows that on an AR(1) covariance its relative variance grows exponentially with the dimension. In practice, bandit algorithms, ranking-and-selection procedures and rating systems approximate these probabilities from the marginal means and variances, or estimate them by counting posterior draws.

The problem is easy to state, and the difficulty is computational. For jointly Gaussian $X = (X_1, \ldots, X_n)$ with independent components, $p_i = \Pr(X_i = \max_j X_j)$ is a one-dimensional

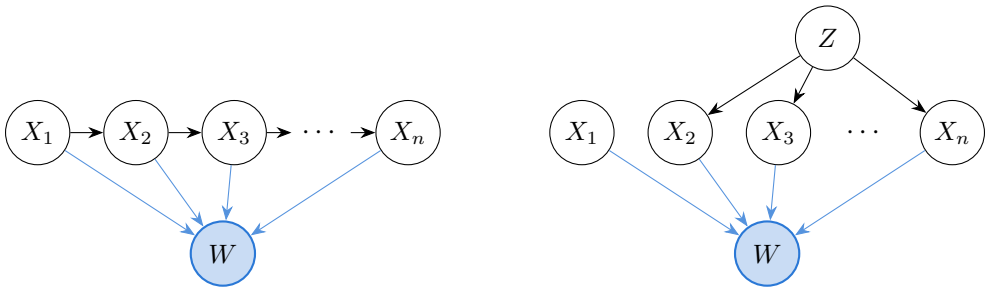
integral, the random-utility representation of Domencich and McFadden (1975). With correlated components it becomes an orthant probability in $n-1$ dimensions, and no earlier method evaluates all n of them deterministically in time linear in n.

We show that for structured Gaussian Bayes nets this can be done. We add to the Gaussian variables a *win node*, the deterministic node $W = \arg\max_i X_i$ with values in $\{1, \dots, n\}$, and ask for what any node in a graphical model must supply: its distribution given its parents, the derivative of that distribution with respect to the parents' means, and the likelihood of an observed value. When the dependence among the X_i runs through a few shared factors, the lattice algorithm of Cotton (2026) for correlated contests gives all n probabilities $\Pr(W = i)$ at once, and Section 3 derives the Jacobian and the exact likelihood of an observed winner from the same computation. When the X_i form a Markov chain, Section 5 computes the same probabilities by a forward and a backward pass over a lattice of values, each pass restricted to paths that stay below a threshold. The same passes give the distribution of the maximum and of the first time the chain crosses a level, and they extend to autoregressions of higher order, whose precision matrices are banded. In both cases the cost is linear in n at fixed numerical resolution.

None of these queries is answered by the univariate marginals alone. Which alternative is best, which sensor reads the extreme, when a monitored series first crosses a threshold, and which entrants make a shortlist all depend on the joint distribution. Section 1.1 gives two networks in which every marginal is the same and the answers are far from uniform.

In applications X is usually a posterior. When $X = \theta \mid \mathcal{D}$ is the belief about the values of n alternatives after seeing data $\mathcal{D}$, p_i is the posterior probability that alternative i is the best, which is also the probability that Thompson sampling chooses i. Section 4 shows that two standard models, simulation under common random numbers and a bandit with a factor prior, keep their posteriors in the factor form of Section 3 after every update, so the win node applies to them without approximation. Section 7 uses it to stop a simulation study, to check it against Thompson sampling on a correlated bandit, and to find the best season of a tennis career.

1.1 Two networks with the same marginals



(a) chain (b) one factor

Figure 1: The two Bayes nets of Section 1.1. Every X_i is $N(0,1)$. A black arrow is a Gaussian conditional distribution; the blue arrows make $W = \arg\max_i X_i$ a deterministic child of every X_i. In (a) the parent of each X_{t+1} is X_t. In (b) the latent Z is the parent of $X_2, \dots, X_n$ and X_1 has no parent.

Figure 1 shows two Gaussian Bayes nets, each with a win node W added as a child of all of its variables. In both, nothing is observed, and the query is the distribution of W: for each i, the

probability p_i that X_i is the largest. Both nets are built so that every X_i is $N(0,1)$. A method that sees only these marginals, for example one that treats the X_i as independent, must give every i the same probability $1/n$. The two nets show how far the true distribution of W can be from that.

In Figure 1(a) the variables form a chain. X_1 has no parent and is $N(0,1)$, and each later variable has its predecessor as its only parent, with $X_{t+1} \mid X_t \sim N(\phi X_t,\, 1-\phi^2)$. This is the stationary AR(1) process. Read X_t as the departure of day t's temperature from its seasonal norm, with ϕ the persistence of weather from one day to the next; the query asks which day of the period is the most unusually hot.

With $\phi = 0.9$ and $n = 20$, the first and last days are each the hottest with probability 0.0974, and a day in the middle with probability 0.0396, against $1/n = 0.05$ from the marginals. A day in the middle has a strongly correlated neighbour on each side, and either of them can beat it. A day at either end has one such neighbour, so it is more often the peak.

In Figure 1(b) the variables share one latent parent. $Z \sim N(0,1)$ is the parent of $X_2, \ldots, X_n$, with $X_i \mid Z \sim N(\sqrt{1-\epsilon}\, Z,\, \epsilon)$, and $X_1 \sim N(0,1)$ has no parent. Read the X_i as a year of returns on n funds: funds $2, \ldots, n$ track the same index Z with tracking variance ϵ, fund 1 follows its own strategy, and the query asks which fund returns the most.

When ϵ is small the index funds rise and fall together, so the contest is close to one between two competitors, fund 1 and the index. As $\epsilon \to 0$, $p_1 \to 1/2$ and each index fund's $p_i \to 1/(2(n-1))$, so the total-variation distance between the true distribution of W and the uniform one tends to $1/2 - 1/n$. With $n = 100$ and $\epsilon = 10^{-4}$, $p_1 = 0.493$ against $1/n = 0.01$. If k independent funds take the place of fund 1, each of them is the best with probability tending to $1/(k+1)$, and the total-variation distance tends to $k/(k+1) - k/n$, which is 0.81 for $n = 100$ and $k = 9$.

2 Related methods

TrueSkill (Herbrich et al., 2007) is already a Bayes net with binary win nodes. Its evidence is a full ranking, whose likelihood factorizes exactly into greater-than factors on the performance differences of adjacent finishers; its approximation is the Gaussian moment matching of the messages. Observing only the winner is different evidence. Section 3.3 gives its likelihood, which is the multinomial probit likelihood. At $n = 2$ it is the binary probit comparison with no draw margin, whose variance is the sum of the two performance variances.

Gaussian message-passing systems and GMRF libraries compute marginals, free energies and samples for very large models, and offer no order-statistic queries. For latent Gaussian models, Bolin and Lindgren (2015) compute excursion sets and exceedance probabilities by sequential importance sampling, and their `excursions` package is the standard tool. On chains and bands the pass of Section 5 computes the same functionals deterministically; for general sparsity, simulation remains the method.

GHK (Geweke, 1989; Hajivassiliou et al., 1996) is the standard method for the orthant probabilities behind p_i, and Cotton (2026) compares it in detail with the factor computation used here. GHK writes one orthant probability as a nested sequence of truncated normal probabilities and simulates it, at $O(n^2)$ per draw for a single alternative.

Within the scope of this paper, factor or chain dependence, the win node improves on it in three respects. It gives all n probabilities in one pass, where GHK needs n runs. Its cost is linear in n. The cost of all n probabilities by GHK grows empirically as $n^{2.8}$ beyond $n = 200$, where Cotton (2026) already measures a gap of two orders of magnitude.

With fixed quadrature nodes it is deterministic: the probabilities and their Jacobian are a reproducible numerical map, with integration error but no simulation noise. Randomized nodes such as scrambled Sobol bring back an error that varies with the scramble, and the logarithm of an unbiased randomized estimate, which is what a log-likelihood needs, is biased downward.

On AR(1) covariances Ridgway (2016) shows the normalized variance of GHK growing exponentially in dimension; we have not compared the two on the chains of Section 6. Outside this scope, for dense high-rank covariance where accuracy is paramount, Cotton (2026) recommends GHK.

For independent components, Menet et al. (2025) write p_i as the one-dimensional integral of Equation (3) without factors (their Equation 2), and their Proposition 1 evaluates it by quadrature on a shared set of integration points with a convergence guarantee. Equation (4) is that evaluator applied conditionally on the factors and averaged over them.

3 The win node on factor-structured parents

This section derives the three parts of the win node when its parents have factor structure: the forward query, which gives all n probabilities $\Pr(W = i)$ (Section 3.1); their derivative with respect to the means (Section 3.2); and the likelihood when the winner is observed (Section 3.3). All three follow from one fact: conditional on the factors, the components are independent.

3.1 The forward query

Let the parents have factor form:

$$X_i \;=\; m_i + b_i' z + \varepsilon_i, \qquad z \sim N(0, I_\rho), \quad \varepsilon_i \sim N(0, d_i), \tag{1}$$

with all the ε_i independent of each other and of z. The covariance is

$$\Sigma \;=\; BB' + \operatorname{diag}(d), \qquad B \in \mathbb{R}^{n\times\rho}, \quad \rho \ll n, \tag{2}$$

where B has rows b_i'. The variables in z are the factors and B holds the loadings. Assume every $d_i > 0$, so the maximum is almost surely unique and the p_i sum to one.

The shared factors make the components dependent. Given z, nothing is shared, and the components are independent with means $m_i + b_i' z$ and variances d_i. The probability that i is the maximum given z is then a one-dimensional integral, and p_i is its average over z. Let $F_{i|z}$ and $f_{i|z}$ be the conditional distribution and density functions of X_i given z. Component i is the maximum when every other component falls below it, so

$$p_i \;=\; \int_{\mathbb{R}^\rho} \phi_\rho(z) \int_{\mathbb{R}} f_{i|z}(x) \prod_{j\neq i} F_{j|z}(x)\, dx\, dz. \tag{3}$$

Component i wins by finishing at some value x with everyone else below it, averaged over the shared factors.

The inner integrands for different i share the product $G_z(x) = \prod_j F_{j|z}(x)$. Dividing out the ith term gives every p_i from the same product:

$$p_i \;=\; \mathbb{E}_z \int f_{i|z}(x)\, \frac{G_z(x)}{F_{i|z}(x)}\, dx. \tag{4}$$

On a lattice of L points in x, with Q quadrature nodes in z, building $\log G_z$ once and subtracting $\log F_{i|z}$ for each i gives all n probabilities in $O(nLQ)$. The accuracy depends on L, on the lattice window and on Q, and the Q needed grows with the number of factors and the geometry of B. The same conditional independence gives top-k membership, with the product replaced by the distribution of the number of components above x. Top-k nodes, with their derivatives, the likelihood of an observed top-k set, and the chain case, are the subject of work in progress.

Only differences between loadings matter. Let $P = I - \mathbf{1}\mathbf{1}'/n$ and write $Bz = (PB)z + \mathbf{1}(\bar{b}'z)$, where $\bar{b}$ is the average row of B. The second term shifts every component by the same amount and changes no ranking, so $p(m, B, d) = p(m, PB, d)$. The number of factor dimensions the integral has to resolve is the rank of PB, which centering can reduce below the rank of B. The exact probabilities are also unchanged by any orthogonal rotation of the columns of B. The numerical properties of the quadrature, including its dependence on orientation, are treated in Cotton (2026).

The lattice in x and the quadrature over z have finite resolution, so the evaluation of Equation (4) has error. Section 6 reports the measured errors.

3.2 The derivative

The derivative of p_i with respect to m_j, for $j \neq i$, is minus the density of the event that i and j tie for the maximum:

$$\frac{\partial p_i}{\partial m_j} \;=\; -\,\mathbb{E}_z \int f_{i|z}(x)\, f_{j|z}(x) \prod_{k \neq i,j} F_{k|z}(x)\, dx, \qquad j \neq i, \tag{5}$$

and $\partial p_i / \partial m_i$ is the sum of the others with the sign reversed, because adding the same constant to every mean leaves p unchanged. The Jacobian is therefore a weighted graph Laplacian whose edge weights are tie densities (Cotton, 2026, §5), and it is positive semidefinite, as it must be: p is the gradient of $\mathbb{E}[\max_i X_i]$ with respect to m, and the expected maximum is convex in m. Its product with a vector costs $O(nLQ)$ on the same lattice, without forming the matrix. Müller et al. (2019) reach the same identity for general joint laws through the surplus function. Equation (5) is the derivative of the exact probabilities. The derivative of the computed lattice probabilities also carries their normalization and, on an adaptive grid, the movement of the grid, so it need not be exactly symmetric or have zero row sums (Cotton, 2026, §§3.2, 5.2).

3.3 The likelihood of an observed winner

A node in a Bayes net must also accept evidence. Condition on $W = k$ and integrate out the factors and the winner's own standardized noise $u \sim N(0, 1)$, independent of z:

$$\Pr(W = k \mid m) \;=\; \mathbb{E}_{z,u} \prod_{j \neq k} \Phi\left(\frac{m_k - m_j + (Bz)_k - (Bz)_j + u\sqrt{d_k}}{\sqrt{d_j}} \right). \tag{6}$$

Given the shared factors and the winner's own performance, each rival independently had to finish below, and each such event is one normal distribution function.

The score of Equation (6), its gradient in m, has a closed form on the quadrature nodes: a weighted sum of Mills ratios, with weights the posterior probabilities of the nodes given the win, computed in one extra pass over the arrays the likelihood already builds. The message an observed win node passes to its Gaussian parents therefore carries exact first-order information.

At $n = 2$ with no factors the likelihood is $\Phi\big((m_k - m_j)/\sqrt{d_k + d_j}\big)$, the binary probit comparison, which is TrueSkill's greater-than factor with no draw margin.

Observing the winner tells less than observing the full ranking, and Equation (6) is the likelihood of that smaller piece of evidence. Moment approximations enter only if the user chooses a Gaussian posterior family, as in expectation propagation. Equation (6) is the multinomial probit likelihood, and with its score it is the estimation core of Cotton (2026, §5).

The implementation used here agrees with finite differences and with dual-number automatic differentiation to 10^{-10}. That checks the differentiation of the implemented numerical likelihood, not its accuracy against the continuum likelihood.

3.4 A quoted price as evidence

The second kind of evidence a win node meets in practice is a price: a market's probability that each alternative wins, quoted before the outcome is known. A price is soft evidence about the parents, and the inverse of the forward query turns it into a Gaussian observation.

Model the market as pricing the same race from its own noisy view of the means, $p = \text{race}(m + \eta)$ with $\eta \sim N(0, \tau^2 I)$, where race is the forward query of Section 3.1 under the market's loadings and noise. The inversion of Cotton (2026, §5) recovers $m + \eta$ from p up to one common level, because prices carry no level: $\sum_i p_i = 1$ fixes $n - 1$ degrees of freedom. The observation therefore lives on the contrast space, $y = P(m + \eta)$ with $P = I - \mathbf{1}\mathbf{1}^\top/n$, and it is linear and Gaussian in m. With a belief $N(\mu, \operatorname{diag} v)$ the posterior precision is $\operatorname{diag}(1/v) + P/\tau^2$, a rank-one correction to a diagonal, and the update is in closed form. The price observes only the contrasts, so what happens to the common level depends on the prior. With a common prior variance the level is independent of the contrasts under the prior and stays where it was: no price can speak to it. With unequal prior variances the level and the contrasts are correlated under the prior, and the level moves. The extreme case shows why. If one entrant's mean is known exactly and another's is not, a price fixes their difference and so fixes the unknown mean, and the average moves with it. The price says nothing about the level; the movement comes from the prior.

The two kinds of evidence compose. Prices are set before the race and the outcome is observed after it, and the two are conditionally independent given the means, so the price update is applied first and the observed-winner message of Equation (6) is applied to the updated belief. In the experiment the belief is kept in the diagonal Gaussian family between messages: the price posterior is exact and its diagonal is carried forward, and the winner message is projected to a diagonal Gaussian as in Section 3.3. The composed row is therefore a moment-matched posterior, not the exact one. Table 1 measures the composition on a seeded simulation: eight alternatives, one shared factor, a market with $\tau^2 = 0.25$, and a prior $N(0, I)$ on the means, averaged over 400 races. The error is the root mean square of the posterior-mean contrasts against the true contrasts; the score is the log probability the posterior predictive assigns to the winner of an independent rerun of the same race.

The market noise τ^2 is the one parameter the model adds, and it is identified by the evidence of the price observations across many races; Cotton (2026, §5) tunes it that way. A market priced under a different model from the outcome, or with a longshot bias, is inverted under its own model and then read as evidence under this one; the inversion is agnostic about where the prices came from.

evidence	RMSE of mean contrasts	mean posterior variance	log score, next winner
none (prior)	0.919	1.000	−2.062
winner only	0.855	0.872	−1.956
price only	0.402	0.300	−1.671
price, then winner	0.399	0.287	−1.656

Table 1: One race, two kinds of evidence. A price with noise $\tau^2 = 0.25$ is worth far more than one observed winner, which is a single draw from the outcome, and the two compose: the posterior after both is tighter and scores better than after either alone. Script: `research/win_nodes/exp1_two_evidence`.

3.5 Places as evidence

Between the winner and the full order lie the results that are actually published: the first k in order with the rest unplaced, the first k as a set, or the bare fact that a named entrant finished among the first k. Each is an event of the win node's relatives, and conditional on the shared factors each has a one-dimensional lattice form, so each passes through the same mixture over factor nodes as Equation (6).

Write f_i and Φ_i for the conditional density and distribution function of entrant i's performance, and $S_i = 1 - \Phi_i$. For the set T of the first k in unknown order, let $G = \prod_{j \notin T} \Phi_j$ be the law of the best outsider and $H = \prod_{i \in T} S_i$ the chance that every member is above x; then

$$\Pr(\text{first } k = T) \;=\; \int G'(x)\, H(x)\, dx \;=\; -\int G(x)\, H'(x)\, dx, \tag{7}$$

and differentiating under either form gives the score as one lattice sum per entrant: nonnegative for a member, nonpositive for an outsider, summing to zero because a common shift of every mean moves nothing. For the ordered prefix $i_1, \ldots, i_k$ with everyone else behind, the last named finishes at x with every unnamed entrant below, $A_k(x) = f_{i_k}(x) \prod_{l \text{ unnamed}} \Phi_l(x)$, each earlier named entrant finishes above the next, $A_j(x) = f_{i_j}(x) \int_{-\infty}^{x} A_{j+1}$, and the probability is $\int A_1$: k cumulative sums on one lattice, for any k. The membership of one entrant among the first k is the top-k query of Cotton (2026) and its Jacobian row is the score. At $k = 1$ all three are the observed winner, and the prefix at $k = n$ is the full order; both reductions are checked to the tolerance of the existing updates, and every message is checked against rejection-sampled posteriors of the generative model.

Table 2 ranks the messages by what they tell. Same simulation as Table 1, 1,600 seeded races, one piece of outcome evidence per race. Who is named carries nothing beyond the message: the entrant reported in the first three is a uniform member of them, and the three whose mutual order is reported are a uniform subset of the field. Naming the winner, or the actual first three, would smuggle a stronger message in; an earlier draft did, and ranked the messages differently. The bare fact that a named entrant finished in the first three is the weakest message; the order among three named entrants, which says nothing about where they finished relative to the rest, is next; the winner alone tells more; the first three as a set tells more than the winner, because it places three entrants rather than one; the prefix adds their order; the full order is the most. Every message's gain over the prior exceeds fifteen paired standard errors across the 1,600 races, from 15 for the named entrant to 39 for the full order.

evidence	RMSE of mean contrasts	mean posterior variance	log score, next wi
none (prior)	0.896	1.000	−2.066
one named entrant in the first three	0.867	0.947	−2.030
order among three named	0.854	0.921	−2.015
winner only	0.830	0.872	−1.967
first three as a set	0.775	0.775	−1.918
first three in order, rest behind	0.759	0.749	−1.891
full order	0.702	0.656	−1.841

Table 2: One race, one outcome message each, ranked by information. Script: `research/win_nodes/exp2_place_evidence`.

4 Posteriors of factor form

The forward query needs the parents in the form of Equation (2). When the parents are a posterior, the question is whether updating preserves that form. Two standard models preserve it exactly after every update: simulation with common random numbers, and a bandit whose arms share features. In both cases the posterior has no more factors than the source of the shared variation, however many alternatives there are, and the $n \times n$ posterior covariance is never formed.

4.1 Simulation with common random numbers

A simulation study compares n systems, such as candidate inventory policies, by running each one many times and averaging its output. Each run is a replicate, and the unknown θ_i is system i's long-run average output. Under common random numbers every system receives the same random inputs in each replicate. A high-demand history raises every policy's profit, so the outputs move together, and the differences between systems are measured more precisely than their levels.

Summarise the shared random inputs of replicate r by ρ variables F_r, such as the demand level and its volatility. The analyst does not record F_r. System i responds to it with known loadings v_i and has its own noise ε_{ir}:

$$Y_{ir} \;=\; \theta_i + v_i' F_r + \varepsilon_{ir}, \qquad F_r \sim N(0, I_\rho), \quad \varepsilon_{ir} \sim N(0, \sigma_i^2). \tag{8}$$

The F_r and ε_{ir} are independent of each other and across replicates, and the noise scales $\sigma_i > 0$ are known. If F_r were recorded, subtracting $v_i' F_r$ from each output would remove the correlation and leave a diagonal posterior. The prior is $\theta \sim N(0, s_0^2 I)$ with $s_0 > 0$.

The covariance of one replicate across systems is $\Lambda = VV' + D$ with $D = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)$. After $n_r \geq 1$ replicates with mean vector $\bar{Y}$, the posterior of θ is Gaussian with precision

$$A \;=\; s_0^{-2} I + n_r \Lambda^{-1}. \tag{9}$$

The Woodbury identity gives

$$\Lambda^{-1} = D^{-1} - D^{-1} V M^{-1} V' D^{-1}, \qquad\qquad M = I_\rho + V' D^{-1} V. \tag{10}$$

With $U = D^{-1}V$ and $S = n_r M^{-1}$, the precision is

$$A = \mathrm{diag}(a) - USU', \qquad a_i = s_0^{-2} + n_r \sigma_i^{-2}, \tag{11}$$

a positive diagonal minus a positive semidefinite term of rank at most ρ. The Woodbury identity again gives

$$A^{-1} = \mathrm{diag}(a)^{-1} + \mathrm{diag}(a)^{-1} U\, C\, U' \mathrm{diag}(a)^{-1}, \qquad C = \left(S^{-1} - U' \mathrm{diag}(a)^{-1} U\right)^{-1}. \tag{12}$$

C is positive definite. Let $K = \left(\begin{smallmatrix} \mathrm{diag}(a) & U \\ U' & S^{-1} \end{smallmatrix}\right)$. The Schur complement of S^{-1} in K is A, which is positive definite, and S^{-1} is positive definite, so K is positive definite. The Schur complement of $\mathrm{diag}(a)$ in K is C^{-1}, so C^{-1} is positive definite too. Hence

$$\Sigma_{\text{post}} \;=\; \mathrm{diag}(d) + BB', \qquad d_i = a_i^{-1}, \quad B = \mathrm{diag}(a)^{-1} U\, C^{1/2}, \tag{13}$$

exactly, for every $n_r \geq 1$.

The posterior mean is $m = A^{-1}h$ with $h = n_r \Lambda^{-1} \bar{Y}$, and both factors are available from Equations (10) and (12). No step approximates. The posterior has factor form because the shared random inputs have factor form.

4.2 A bandit whose arms share features

In a bandit, a decision maker repeatedly picks one of n arms and observes a noisy reward. Here θ_i is arm i's mean reward, and pulling arm a returns $\theta_a + N(0, \sigma_n^2)$ with $\sigma_n > 0$. Arm i has known features v_i, and $\theta_i = v_i'\beta + e_i$ with a shared $\beta \sim N(0, I_\rho)$ and independent $e_i \sim N(0, d_{0i})$, $d_{0i} > 0$. The prior is $\theta \sim N(0, \Sigma_0)$ with $\Sigma_0 = VV' + \mathrm{diag}(d_0)$, which has factor form.

Linear Thompson sampling (Agrawal and Goyal, 2013), the version Chapelle and Li (2011) evaluated on display advertising and news recommendation, is this model without the residual, with factor rank equal to the number of features. A Gaussian-process prior with a kernel of finitely many features and an independent residual has this form exactly. A dense kernel does not, and an inducing-point approximation to one has a dense residual (Titsias, 2009), so for those a diagonal residual is an added approximation.

With per-arm pull counts c, the posterior precision is $A = \Sigma_0^{-1} + \sigma_n^{-2}\,\mathrm{diag}(c)$. Equation (10) applied to Σ_0 puts A in the form of Equation (11), with $a_i = d_{0i}^{-1} + c_i \sigma_n^{-2}$, $S = M_0^{-1}$ and $M_0 = I_\rho + V' \mathrm{diag}(d_0)^{-1} V$, and the rest of Section 4.1 applies unchanged. The posterior is exactly factor plus diagonal after every pull, with no more factors than the prior.

The same posterior has a form with no subtraction inside an inverse. Let $\lambda_i = c_i/\sigma_n^2$ be the observation precision accumulated on arm i, $r_i = 1/(1 + d_{0i}\lambda_i)$, $R = \mathrm{diag}(r)$ and $H = I_\rho + V' \mathrm{diag}(\lambda_i r_i) V$. Then

$$\Sigma_{\text{post}} \;=\; \mathrm{diag}(d_{0i} r_i) + RVH^{-1}V'R, \qquad B_{\text{post}} = RVH^{-1/2}. \tag{14}$$

H is the identity plus a positive semidefinite matrix, so it is positive definite. An arm never pulled has $\lambda_i = 0$, and arms observed with different noise levels enter through their own λ_i. Equation (14) agrees with direct inversion of the dense precision to machine precision, and is the form to implement.

5 The win node on chain and band parents

Now let the precision matrix of X be banded with bandwidth b, so that X is Markov of order b; the Gauss–Markov chain is $b = 1$. Write the chain as $X_1 \sim N(\mu_1, \sigma_0^2)$ and $X_{t+1} \mid X_t \sim N(\mu_{t+1} + \phi_t(X_t - \mu_t), s_t^2)$. A tridiagonal precision matrix gives these parameters directly. Eliminating the variables from the last to the first gives a bidiagonal factor whose rows are the forward regressions of X_{t+1} on X_t; an ordinary upper Cholesky factor gives the same regressions running backwards. Discretize the state on a lattice of L points, as in Section 3.1, so that the transition kernels become $L \times L$ matrices T_s and the passes below are approximations to the corresponding integral operators.

Law of the maximum and first passage. The maximum stays below u when every step does, so restrict each transition to values below u. With T_s the lattice transition from X_s to X_{s+1} and π_u the initial mass vector cut at u,

$$\Pr(\max_{s \le t} X_s \le u) \;\approx\; \mathbf{1}^\top T_{t-1}^{(u)} \cdots T_1^{(u)} \, \pi_u, \qquad T_s^{(u)}(x' \mid x) = T_s(x' \mid x)\, \mathbf{1}\{x' \le u\}, \tag{15}$$

with equality for the continuous operators. This is one forward pass. A banded precision does not make the lattice transition matrix banded, and with dense $L \times L$ transitions the pass costs $O(nL^2)$ per threshold.

The mass lost at step t of the same pass is the probability that the chain first crosses u at time t, so the pass also gives the first-passage law. The expected maximum M follows from its distribution function: on a threshold window $[a, b]$ that holds all but a negligible part of its mass, $\mathbb{E}[M] \approx a + \int_a^b [1 - F_M(u)]\, du$. The rectangle rule for this integral has an error of order Δu, measured at $+0.053$ on a ten-node chain, and the trapezoid rule removes that leading error.

A hard cut of the lattice at u has error $O(\Delta x)$; giving the boundary cell its fractional occupancy restores $O(\Delta x^2)$, which improves agreement with the i.i.d. closed form at 400 lattice points from 10^{-2} to 2×10^{-4}.

Argmax probabilities. By the Markov property, given $X_t = u$ the path before t and the path after t are independent, so

$$\Pr(X_t = \max_s X_s) \;=\; \int f_t(u)\, \Pr(X_1, \ldots, X_{t-1} < u \mid X_t = u)\, \Pr(X_{t+1}, \ldots, X_n < u \mid X_t = u)\, du, \tag{16}$$

where f_t is the density of X_t. On the lattice the integral becomes a sum over thresholds u_k, and for each u_k a forward pass and a backward pass restricted below u_k give the two conditional probabilities for every t at once. Near-ties matter here: a hard cut at u_k discards the mass of paths whose maximum falls in the same lattice cell as X_t, and the probabilities then sum to less than one. Giving that cell weight one half removes the first-order loss. With dense transitions each threshold costs $O(nL^2)$, and all n probabilities cost $O(nL^3)$.

Bandwidth b by state lifting. The vector $(X_t, \ldots, X_{t+b-1})$ is Markov, so the same restricted passes run on the lifted state at $O(nL^{b+1})$ per threshold: linear in n and exponential only in the bandwidth. This covers the law of the maximum and first passage. The argmax pass on the lifted state is not implemented. At $b = 2$ the lift covers AR(2), including the cyclical case with complex roots that has no $b = 1$ analogue. With the second coefficient set to zero the lift reproduces the $b = 1$ pass to 5.6×10^{-16} on the same grid.

6 Accuracy

All numbers in this section come from committed, seeded scripts in the public repository. Monte Carlo is used only as a check.

Factor parents. The accuracy of the factor computation is studied in Cotton (2026). On ten posteriors from the experiments of Section 7, raising the lattice from 257 to 2,049 points changes no entry by more than 4×10^{-16}, and replacing the default factor rule by 2^{15} scrambled-Sobol nodes changes no entry by more than 1.2×10^{-6}. Four million Monte Carlo draws differ from the computed probabilities by 1 to 4×10^{-4} per entry, which is their own standard error. No error bound is proved. The script is `research/rs_crn/exp3_quadrature`.

Chain and band parents. The passes of Section 5 are checked against simulation and against an exact law; the scripts are `research/tridiagonal/exp1` to `exp4`. On the AR(1) chain of Section 1.1, 4×10^5 simulated paths agree with the computed argmax probabilities to within 8.1×10^{-4}, about twice their standard error. For a random walk the finite-n argmax law is known exactly: $p_t = q_{t-1} q_{n-t}$ with $q_k = \binom{2k}{k} 4^{-k}$ (Sparre Andersen, 1954). At $n = 40$ the lattice pass is within 1.1×10^{-4} of it, a relative error of at most 0.23%, while the continuum arcsine law is 0.011 away at the ends. The argmax probabilities sum to 0.9996 for the AR(1) chain and 0.998 for the walk; the shortfall is lattice error and is not renormalized away. The same pass handles drifted, barriered and mean-reverting walks, which have no closed form.

Rare thresholds. At a low threshold the law of the maximum is an orthant probability, $\Pr(X_1 \le u, \dots, X_n \le u)$, and as u falls it becomes a rare event. Plain posterior simulation needs about $1/p$ draws to see it at all. GHK does not, because it samples from truncated proposals and weights them, and it is not compared here. The restricted pass costs the same whatever p is, 136 ms per threshold in Table 3.

u	lattice, L=400	lattice, L=800	10^6 draws	draws for 10% error
0.0	6.471×10^{-3}	6.465×10^{-3}	6.432×10^{-3}	1.5×10^4
-1.0	1.756×10^{-5}	1.753×10^{-5}	1.60×10^{-5} (16 hits)	5.7×10^6
-1.5	3.088×10^{-7}	3.082×10^{-7}	no hits	3.2×10^8
-2.0	2.531×10^{-9}	2.525×10^{-9}	no hits	4.0×10^{10}
-2.5	9.464×10^{-12}	9.436×10^{-12}	no hits	1.1×10^{13}
-3.0	1.588×10^{-14}	1.582×10^{-14}	no hits	6.3×10^{15}

Table 3: $\Pr(\max_t X_t \le u)$ for a stationary AR(1) chain with $n = 50$ and $\phi = 0.9$.

Simulation agrees where it has hits and returns zero where it has none. Below $u = -1.5$ there is no simulation check, and the evidence for the computed values is that they are stable under refinement of the lattice: the two lattices agree to within 0.4% at every threshold, including 1.58×10^{-14}.

7 Applications

The four applications use posteriors of the two structures. The first three concern the probability that each of several alternatives is the best under a factor-form posterior: Section 7.1 relates it to Thompson sampling and to two approximations in current use, Section 7.2 uses it to stop a simulation study, and Section 7.3 checks it against Thompson sampling on a correlated bandit.

The fourth, Section 7.4, asks which season of a tennis career was the best, under a chain posterior fitted to match results.

7.1 Thompson sampling probabilities and their approximations

With what probability will Thompson sampling show each of fifty advertisements next? Let $X = \theta \mid \mathcal{D}$ be a posterior over the values of n alternatives. Thompson sampling draws one sample $\tilde{\theta}$ and chooses the alternative with the largest sampled value, so it chooses i with probability $p_i = \Pr(W = i)$.

Lemma 8 of Tarbouriech et al. (2023) states the general version: the expected fraction of pulls each arm receives under Thompson sampling equals its posterior probability of being best. Choosing i with the computed probability $\hat{p}_i$ is therefore Thompson sampling up to the evaluation error of Section 6. The policy is still randomized. Only the evaluation of its probabilities is deterministic.

Thompson sampling itself never needs the numbers p_i. A stopping rule needs $\max_i p_i$. A shortlist needs the k entries most likely to be best, an allocator needs to know which p_i are close, a constraint on how often any arm may be shown needs the probabilities it constrains, and off-policy evaluation needs them as propensities. In each of these uses they are currently approximated, either by counting posterior draws or from the marginal means m_i and standard deviations $s_i = \sqrt{d_i + \|b_i\|^2}$. Two marginal approximations are compared below.

F-LITE (Menet et al., 2025) treats the alternatives as independent, $\prod_i N(m_i, s_i^2)$, and approximates the probability of being best by $\Phi((m_i - \kappa^*)/s_i)$, with a threshold κ^* chosen so that these sum to one. The threshold step is a second approximation. With two independent arms, $N(1,1)$ and $N(0,1)$, it gives 0.691 for the first arm, against the exact 0.760.

The *VBOS policy* is the variational Bayesian optimistic sampling rule of O'Donoghue and Lattimore (2021), extended by Tarbouriech et al. (2023). In the near-closed form of Menet et al. (2026) it is

$$\tilde{\pi}_i \;=\; v\left(\frac{m_i - \kappa^*}{s_i}\right), \qquad v(c) = \exp\Big(-\tfrac{1}{8}\big(\sqrt{c^2+4} - c\big)^2\Big), \tag{17}$$

with κ^* again chosen so the entries sum to one. The VBOS policy departs from p on purpose, and its regret analysis depends on that departure. Its distance from p matters because Menet et al. (2026) use it as the trainable substitute for p.

Both approximations ignore correlation, and the second example of Section 1.1 shows how large the resulting error can be. A simpler example fixes the mean gap and varies the size of a shared shock. Let $\theta_1 = 3 + Z + \varepsilon_1$ and $\theta_2 = Z + \varepsilon_2$, with $Z \sim N(0, \tau^2)$ and independent $\varepsilon_i \sim N(0,1)$. The shared Z cancels from the difference, so $p_1 = \Phi(3/\sqrt{2}) \approx 0.983$ for every τ. F-LITE sees only the marginal variances $1 + \tau^2$ and gives $\Phi\big(3/(2\sqrt{1+\tau^2})\big)$, which is 0.933 at $\tau = 0$, 0.682 at $\tau = 3$, and tends to one half as τ grows.

7.2 Stopping a simulation study

Has a simulation study run long enough to say which of a hundred inventory policies is best? The first experiment runs the simulation study of Section 4.1 and asks when it can stop. A study stops when it is confident enough that it has found the best system. The question is whether a stopping rule that uses the computed probabilities stops earlier than rules that use the marginal approximations, and whether it still selects the best system as often as its confidence level promises.

The setting is the posterior of Section 4.1 with $n = 100$ systems, $\rho = 2$ shared input streams and prior scale $s_0 = 1$. True means are drawn from the prior. In every regime each system's replicate output has variance one, and the *factor share* is the fraction of that variance that comes from the shared inputs. The posterior uses the outputs of all n systems jointly, so its marginal variances still differ between regimes: after 20 replicates each θ_i has posterior variance 0.048 in the independent regime and 0.027 in the aligned one. The comparison is between rules within a regime.

Correlation can mislead a marginal-only rule in two directions. If the leading systems move together, their difference is less uncertain than their marginals suggest, and the rule is too cautious. If they move oppositely, their difference is more uncertain, and the rule is too confident. The regimes produce each case, with an uncorrelated control.

- *Independent*: factor share zero. This is the control.
- *Aligned*: every system loads on the same stream, with factor share 0.6. Differences between systems then vary less than their marginal variances suggest. Every row of B is the same, so $PB = 0$ and the computed probabilities are those of an independent contest, which the marginal rules do not know. This is the shared shock example of Section 7.1 with a hundred systems.
- *Opposed*: half the systems load positively and half negatively on one stream, with factor share 0.6. Differences between systems in opposite halves then vary more than their marginal variances suggest.

Three stopping rules differ only in how they estimate p: the win node of Section 3, F-LITE and the VBOS policy. After every replicate a rule recomputes its estimate, and it stops the first time the largest entry reaches $1 - \delta$, with $\delta = 0.05$, selecting that entry. A fourth rule, drawing M samples from the correlated posterior and counting argmax frequencies, uses the correlation and costs $O(Mn\rho)$. At $p = 0.95$, $M = 500$ draws give a standard error of 0.0097, and a 95% margin of 0.01 needs about 1,800. It was not run.

With the true probabilities the rule has a Bayesian guarantee. Let τ be the first time $\max_i p_i \geq 1 - \delta$, let B_r be the budget of replicates, and on $\{\tau \leq B_r\}$ let I be the selected alternative and I^* the best. Then

$$\Pr(I \neq I^* \mid \tau \leq B_r) \;=\; \mathbb{E}\big[1 - p_I(\mathcal{D}_\tau) \mid \tau \leq B_r\big] \;\leq\; \delta. \tag{18}$$

Optional stopping does not affect this statement, which averages over the prior. It is not a frequentist guarantee for a fixed θ.

For computed probabilities with $\|\hat{p} - p\|_\infty \leq \eta$ at stopping, the error is at most $\delta + \eta$, and a threshold of $1 - \delta + \eta$ restores δ. At the ten posteriors measured in Section 6, $\|\hat{p} - p\|_\infty$ was about 10^{-6}. That is a measurement at those posteriors and not a bound at every stopping time.

The threshold is nominal, and its true confidence level differs by rule. With two independent arms of equal posterior standard deviation s, an entry of 0.95 requires a mean gap of $2.33\,s$ for the exact probabilities, $3.29\,s$ for F-LITE and $4.84\,s$ for the VBOS policy. The true posterior probability of a correct selection at those gaps is 0.950, 0.990 and 0.9997. Part of any extra sampling by the marginal rules is this higher effective target.

In Table 4, every rule's realized PCS among its stopped runs is above nominal. The lowest, the win node rule's 0.970 in the opposed regime, has 167 stopped runs and a standard error of 0.013.

regime	rule	mean replicates	realized PCS	censored of 200
independent	win node	126.5	0.971	29
	F-LITE	188.7	1.000	57
	VBOS	252.9	1.000	88
aligned	win node	81.2	0.973	17
	F-LITE	175.2	1.000	53
	VBOS	236.7	1.000	90
opposed	win node	122.7	0.970	33
	F-LITE	188.6	0.993	64
	VBOS	244.2	1.000	92

Table 4: Stopping at nominal 95% confidence. Each regime has 200 replications and a budget of 400 replicates. Mean replicates is the mean of the stopping time capped at 400, so a censored run counts 400. Realized PCS is the fraction of stopped runs that selected the true best system. The stopped runs differ by rule.

At this common nominal threshold and budget, the two marginal rules used 1.5 to 2.9 times the capped mean replicates of the win node rule, and hit the budget far more often. The gap is largest in the aligned regime, where the win node rule stops after 81.2 replicates on average, against 175.2 and 236.7, and stops in 183 of 200 runs, against 147 and 110. The thresholds are nominal and not calibrated to a common PCS, so these ratios compare spend at a common threshold. They do not compare spend at a matched true error.

The marginal estimates are far from the computed probabilities along the way. Before stopping in the aligned regime, F-LITE is at median total-variation distance 0.116 from them, with ninetieth percentile 0.174, and the VBOS policy is at 0.133, with ninetieth percentile 0.207. In the independent regime the distances are 0.063 and 0.095, because F-LITE's threshold step is biased even for an independent posterior and the VBOS policy departs from p by design. At the experiment's Monte Carlo checks, using 2×10^5 draws at twelve posteriors, the largest total-variation distance of the computed probabilities was 0.0026, which is the noise of the check.

7.3 A bandit with correlated arms

Does choosing arms from the computed probabilities reproduce Thompson sampling when the arms are correlated? On the bandit of Section 4.2, with $n = 50$ arms, horizon 200 and 100 replications, choosing arms with the computed probabilities gives cumulative Bayes regret within two percent of Thompson sampling's, in an independent regime and in one where each half of the arms shares a factor carrying 0.75 of the prior variance. The paired differences, -3.0 ± 12.0 and $+1.9 \pm 8.0$ at 95%, are consistent with zero, which is what Lemma 8 requires. Used as the policy, the VBOS approximation has ten to twelve percent more regret than Thompson sampling, a paired difference of 16.1 ± 10.6 and 14.7 ± 7.4, with its action distribution at total-variation distance 0.13 to 0.16 from the computed probabilities. F-LITE has three to six percent more, 5.7 ± 11.3 and 6.8 ± 7.2, at total-variation distance about 0.06. The script is `research/rs_crn/exp2_bandit`.

7.4 Peak seasons in ATP tennis

Which year was the most likely to be Nadal's best? A year was his best if his ability that year exceeded his ability in every other year, so the question is about the joint law of his ability across seasons. A rating system reports a yearly estimate with an error bar, and those marginals do not answer it.

A rating model is a Bayes net whose evidence is observed win nodes, and it produces a chain posterior of the kind Section 5 reads. Give one player a latent ability path $\theta_1, \dots, \theta_T$ across the years of a career and put a random-walk prior on it. Each match against an opponent of ability a_o contributes $\log \Phi(\pm(\theta_t - a_o))$, which is Equation (6) at $n = 2$, and touches one year, so the negative Hessian of the log likelihood is diagonal, with positive entries. The prior's precision is tridiagonal, the Laplace posterior precision is their sum, and the career posterior is a Gauss–Markov chain.

The data are 129,559 ATP matches from 1985 to 2024 and the 1,453 players with at least fifteen of them. Static abilities for the field come from a MAP fit of the probit Bradley–Terry likelihood. For one player at a time, that constant is replaced by a yearly path, fitted under the random-walk prior with the innovation scale chosen by the Laplace marginal likelihood. Opponents stay at their static strength, so the fit is conditional on the rest of the field.

A rating table reports the year with the highest fitted ability. Equation (16) gives the probability that each year was the best. Table 5 shows both.

player	fitted peak	Pr(that year)	80% credible set
Federer	2006	0.52	2005–2006
Nadal	2013	0.27	six years within 2008–2019
Djokovic	2015	0.43	2013–2015
Sampras	1994	0.47	1994–1996
Agassi	1995	0.55	1995, 2002–2003
Murray	2016	0.59	three years within 2009–2016

Table 5: The year of highest fitted ability, the posterior probability that it was the best year, and the smallest set of years holding 80% of that probability.

The fitted peaks are the years a reader would expect, and the probabilities show how uncertain they are. Federer's 2006 carries 0.52 against 0.41 for 2005. Nadal's most likely peak year carries 0.27, and his law is bimodal: 0.267 at 2013 and 0.220 at 2018, with little mass on the years between. Agassi's credible set is not an interval. It holds 1995 and the 2002–2003 resurgence and excludes the years in between. That is a property of the joint law, and no year's marginal shows it.

Figure 2 puts the two on one time axis. A year with fewer matches has a wider posterior, and that can give it more argmax mass, though it need not: in a chain a wider marginal can also have less. In this fit, Nadal's 2010 has posterior mean 2.042 from 80 matches and probability 0.066, and his 2012 has mean 2.051 from 48 matches and probability 0.088. The two years also differ in their neighbours and correlations, so the comparison does not isolate the effect of the number of matches.

Two hundred thousand draws from each posterior agree with the computed probabilities to 0.0020 or better, across all six careers and every year. The Gaussian posterior is a Laplace approximation, and the probabilities are computed for that posterior. The script is `exp6_career_peak`

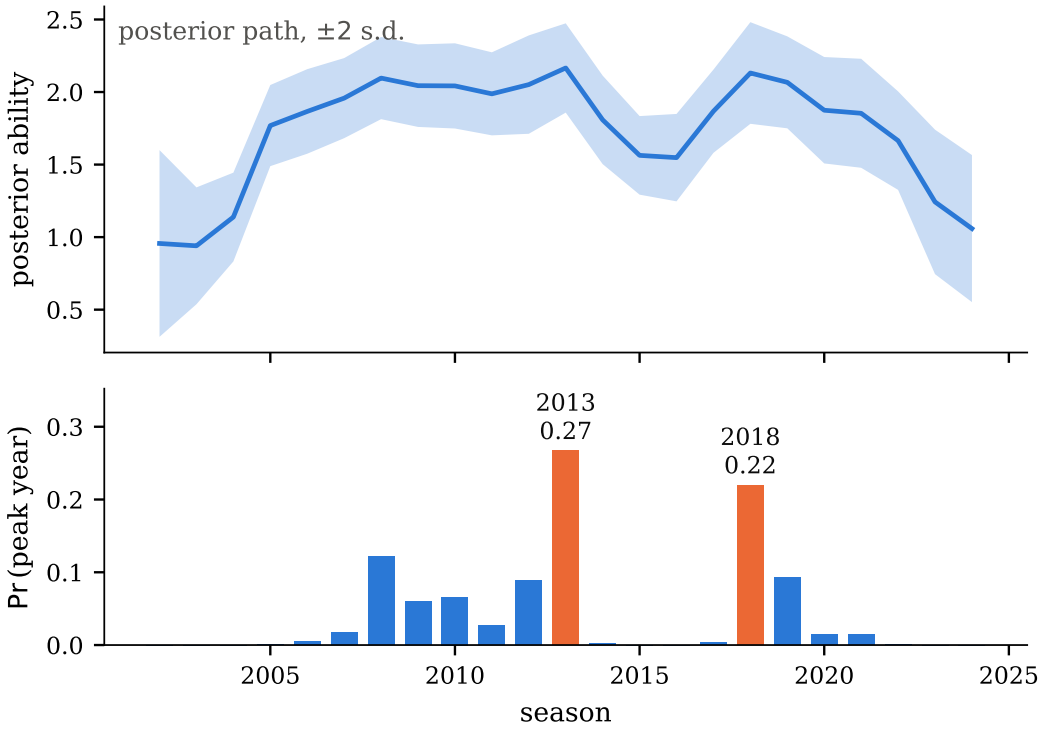


Figure 2: Rafael Nadal's posterior ability path from 1,317 matches (top) and the probability that each year was his best (bottom). The path is one broad plateau, and the argmax law is bimodal, at 2013 and 2018, with almost no mass on the seasons between them. Neither the band in the top panel nor any summary of the yearly marginals shows this.

under `research/tridiagonal`.

8 Software

The factor win node, its top-k relatives, both Jacobian blocks, the calibration inverses, and the evidence messages of Sections 3.3–3.5 behind one front door (`update_race`: prices, winner, order, prefix, top set, membership) are in the `winning` package of Cotton (2026): a Python reference with R, JavaScript, Julia and Rust ports, each checked against the same embedded reference vectors, most to 10^{-15}. The observed win node, Equation (6) with its analytic score, is the estimation core of `MultinomialProbit.jl`, whose score is also checked by forward-mode automatic differentiation through the same code. The chain and band layer is `GMRFExtremes.jl`, which reads a Gauss–Markov chain off a tridiagonal precision and dispatches on GMRF library types. Wider bandwidth and general sparsity are refused, with a pointer to the lifted pass and to Bolin and Lindgren (2015) respectively.

9 Scope

Structure. The reduction requires structure. Factor dependence of rank ρ costs a Q-node quadrature and is exact to quadrature accuracy. On a chain the argmax probabilities cost $O(nL^3)$

with dense transitions. For precision bandwidth b the law of the maximum and first passage cost $O(nL^{b+1})$ per threshold, which is practical for $b \leq 3$. General sparse precision gets neither treatment, and the excursion functionals then belong to the sampling methods of Bolin and Lindgren (2015). The argmax pass on the lifted state at $b \geq 2$, and a pass cheaper than $O(L^{b+1})$ for b beyond about 3, are left for future work.

Rank. The error of the marginal approximations can approach one half with every marginal held fixed, whatever the factor rank. The comparison with counted posterior draws depends on rank. On the posteriors of the two experiments, where the centered loadings have rank at most two, one evaluation for a hundred alternatives takes about ten milliseconds and is accurate to 10^{-6} per entry.

Counting draws for ten milliseconds gives about twenty thousand draws, which estimate an entry near 0.01 to seven percent of itself. A standard error of 10^{-6} on an entry near 0.01 would take about 10^{10} draws.

At factor ranks three to five, on single hundred-arm posteriors of a linear sampler, one evaluation takes 0.3 to 1 second and 200,000 draws take 0.3 second, with errors of the same order. The win node is faster than counting at low rank, and the rank that matters is that of the centered loadings.

Exactness of the posteriors. The derivations of Section 4 are exact because the observations are generated with factor structure. When shared inputs enter a simulation nonlinearly, a low-rank model of Λ is an approximation, and the posterior inherits its error. A dense covariance can be fitted to factors, blocks, nested groups or a tree on intake (Cotton, 2026). The fit introduces model error, and its effect on the probabilities depends on the covariance of the contrasts and its conditioning: a small covariance residual can still move a nearly deterministic comparison a long way.

Allocation. The probabilities say how uncertain the choice of best is. They do not say where the next measurement should go. The knowledge gradient (Frazier et al., 2008) answers that question for one objective, the expected improvement in the value of the final choice, usually the largest posterior mean, and Tuisov (2026) treats it as the myopic value of computation. Reducing the entropy of p, which is entropy search, and raising $\max_i p_i$, which aims at correct identification under zero-one loss, are different objectives. Hennig and Schuler (2012) note that both are objectives of an optimizer. A bandit is scored on the values it collects.

Entropy search. On a finite set of representer points, the probability that the minimum of a Gaussian process lies at each point is p in the minimum convention. Hennig and Schuler (2012) approximate it by expectation propagation at cost $O(n^4)$, and their implementation uses fifty representer points. For a belief of factor form the win node gives the same probabilities in time linear in n, and the closure of Section 4 is the case in which the hypothetical posterior after a measurement stays in factor form. Entropy search also uses derivatives with respect to the covariance, which the win node does not supply, and this paper does not run it.

Top-k nodes. The win node is the $k = 1$ case of a top-k node. For factor parents the forward query and its derivatives are implemented in the `winning` package; the full treatment, including chains and observed top-k sets, is work in progress.

Off-policy evaluation. The replay method of Li et al. (2011) needs a uniformly random log, because the sampler's own propensities were not available, and discards every logged event whose random arm differs from the policy's choice. With the propensities computed, the log of a correlated Thompson sampler can be used directly. This paper does not test that.

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